1 Reaction to sRFC Errors

Usage

With sRFC-communication the calling system will receive next to a technical return code also business information regarding the success of an intended action.

The evaluation of the return parameters from the SAP function module can be optimized here as regards the question whether to reject specific data immediately (in this case the next data record would be edited) or whether to wait "online" for a couple of seconds before trying again to post the data to SAP.

Requirements

You use the synchronous RFC for the communication with SAP.

You execute the upload client hysapupl.exe/out by the command parameter /SINGLE_IDOC

Mapping

A general error handling function is integrated into the confirmation program to SAP hysapupl.exe/out that provides for a uniform error handling of sRFC-calls. This function assumes the error handling:

- if there is an error such as a RFC-exception and/or an exception of the function module,
- if the posting success is returned to a BAPIRET-structure,
- If the posting success is returned to a BAPIRET-table,
- if in case of the QM-IDI the posting success is returned to a QIERR-table.
- This uniform return code handling is made synchronously for all called RFC-modules, and in doing so existing island solutions were customized to this uniform processing method.

Processing is activated using the new program parameter/ RET_CODE_EVALUATION. This is used to evaluate the existing parameter/ REDO_CANCEL= for all synchronous calls. If this is not set, HKMPP-PDCC will be used as default. In addition, it will be checked in the sap_fb_return_cfg table whether it contains entries for the respective synchronous module.

In error handling the existing table sap_fb_return_cfg will be evaluated and any returned error will be handled correspondingly. To do so, the following logics is used:

PRIO 1 :

Error handling when the function module returns an exception.

This will be checked against the table sap_fb_return_cfg and be handled according to the configuration applying to that combination of function module name and exception.

If there is no entry, the data record will "normally" be handled as REDO-cancel and will automatically be set to "ToDo" and/or "DONE_ERROR".

PRIO 2 :

Error handling when the functional module does NOT return an exception but return the result in a return structure (Prio 2).

This will be checked against the table sap_fb_return_cfg and be handled according to the configuration applying to that combination of function module name and ret_type / ret_id / ret_number.

Once the processing will be completed, the data record status of the des ret_type in the return structure will be implemented according to the following pattern:

- "S" DONE
- "W" DONE
- "I" DONE
- "E" DONE ERROR
- "A" DONE ERROR

PRIO3 :

Error handling when the functional module does NOT return an exception but return the result in a return table.

This return table may include more entries of different ret_type. This is the reason why this table will be used to iterate and in line with the following priority the table sap_fb_return_cfg will be searched for an entry:

- ret_type = A
- 2. ret_type = E
- 3. ret_type = W

If an entry will be found, the configuration in the table sap_fb_return_cfg will be used for further handling.

Once the processing will be completed, the data record status of the des ret_type in the return structure will be implemented according to the following pattern:

- "S" DONE
- "W" DONE
- "I" DONE
- "E" (at least one entry – Prio 2) DONE ERROR
- "A" (at least one entry – Prio 1) DONE ERROR

PRIO4 :

Error handling when the functional module does NOT return an exception but return the result in a QUIERR table.

This return table may include more entries of different ret_type. This is the reason why this table will be used to iterate and in line with the following priority the table sap_fb_return_cfg will be searched for an entry:

- ret_type = A
- 2. ret_type = E
- 3. ret_type = W

If an entry will be found, the configuration in the table sap_fb_return_cfg will be used for further handling.

Once the processing will be completed, the data record status of the des ret_type in the return structure will be implemented according to the following pattern:

- "S" DONE
- "W" DONE
- "I" DONE
- "E" (at least one entry – Prio 2) DONE ERROR
- "A" (at least one entry – Prio 1) DONE ERROR

If an entry is found in the table sap_fb_return_cfg for a return code, the system will interpret this entry as follows:

- ACTION = "A“ (only for SAP-PPPDCC)
  The data record will be marked immediately as DONE_ERROR without taking the program parameter "/REDO_CANCEL" into account.

- ACTION = "R"
  The system will wait for the time period specified in DELAY (in seconds) online (with live connection to SAP) and once this time period has elapsed will attempt to post the data once again.
  If this is possible (=successful), the data record will be marked as DONE.
  If this can again not be posted (e.g. because the data record is still being blocked in SAP), it will be marked in HYDRA as TODO (try again) or DONE_ERROR (impossible to post) while the value defined via the program parameter "/REDO_CANCEL" will be taken into account. If REDO_CANCEL is not explicitly specified during the program call, the DEFAULT value will be used.

Entries or changes to this table need to be agreed with MPDV since erroneous configurations may lead to unwanted side effects.

Table: sap_fb_return_cfg

Field	KEY	Type	L	D	Meaning	Notes/ Usage in HYDRA
FB_NAME	X	CHAR	30		Name of the functional module	Used to determine whether there are entries in this table to this FB
EXCEPTION	X	CHAR	30		Exception	Exception that is triggered by the module FUTURE USE
RET_TYPE	X	CHAR	1	0	Message type: S Success, E Error, W Warning, I Info, A Abort	Possible entries: "E" --> Error "A" --> Abort "W" --> Warning
RET_ID	X	CHAR	20	0	Messages, message class	Message type, e.g. "RU"
RET_NUMBER	X	NUMC	3	0	Messages, message number	Message number, e.g. "486"
RET_MESSAGE		CHAR	220	0	Message text	FUTURE USE
RET_LOG_NO		CHAR	20	0	Application log: Protocol number	FUTURE USE
RET_LOG_MSG_NO		NUMC	6	0	Application log : internal serial number of the message	FUTURE USE

Field	KEY	Type	L	D	Meaning	Notes/ Usage in HYDRA
RET_MESSAGE_V1		CHAR	50	0	Messages, message variable	FUTURE USE
RET_MESSAGE_V2		CHAR	50	0	Messages, message variable	FUTURE USE
RET_MESSAGE_V3		CHAR	50	0	Messages, message variable	FUTURE USE
RET_MESSAGE_V4		CHAR	50	0	Messages, message variable	FUTURE USE
RET_PARAMETER		CHAR	32	0	Name of the parameter	FUTURE USE
RET_ROW		INT4	10	0	Line in parameter	FUTURE USE
RET_FIELD		CHAR	30	0	Field in parameter	FUTURE USE
RET_SYSTEM		CHAR	10	0	System (logical system) that issued the message	FUTURE USE
ACTION		CHAR	2		Action to be executed	This defines which action will be executed by hysapupl.exe/out: "A" --> Cancel (immediate setting apart) "R" --> Online repetition
DELAY		NUMC	8		Delay in seconds	If "R"epeat is stored to the ACTION field, than this defines the time interval in seconds after which the repetition is to be made.